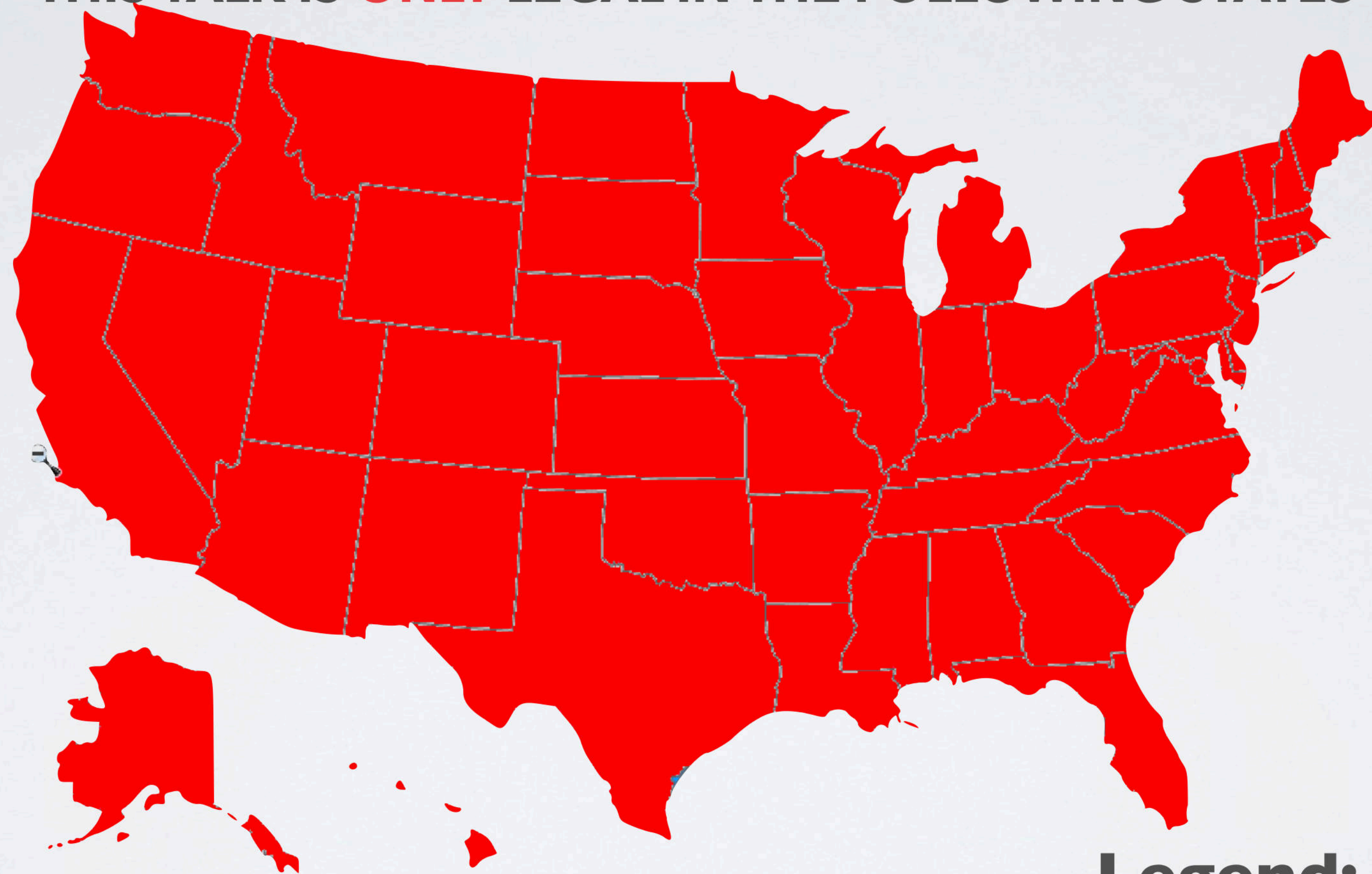




Bill Swearingen / @hevnsnt

WARNING

THIS TALK IS **ONLY** LEGAL IN THE FOLLOWING STATES



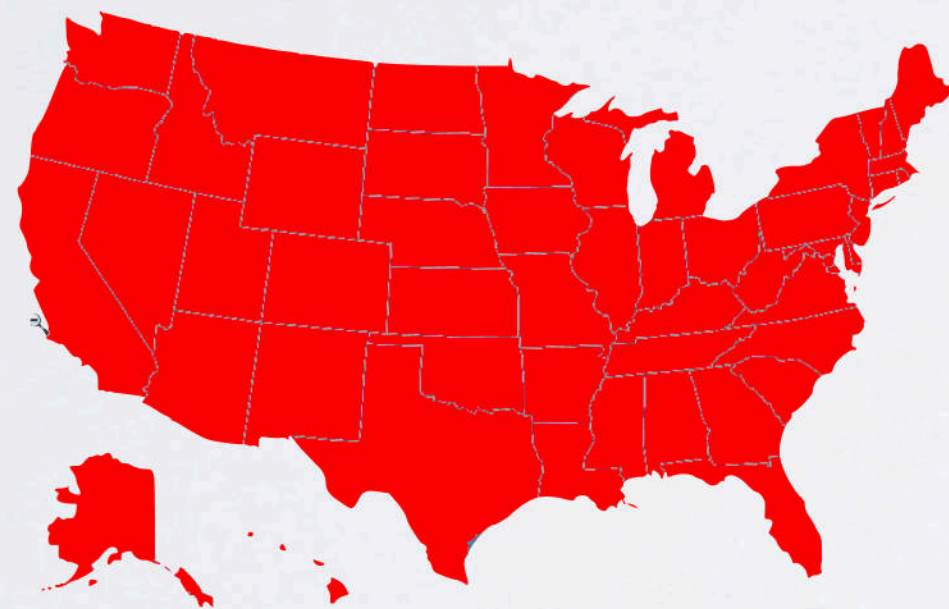
Legend:

RED ILLEGAL

GREEN LEGAL

WARNING

THIS TALK IS **ONLY** LEGAL IN THE FOLLOWING STATES



Legend:

RED ILLEGAL

GREEN LEGAL

This talk is rated **R** for:

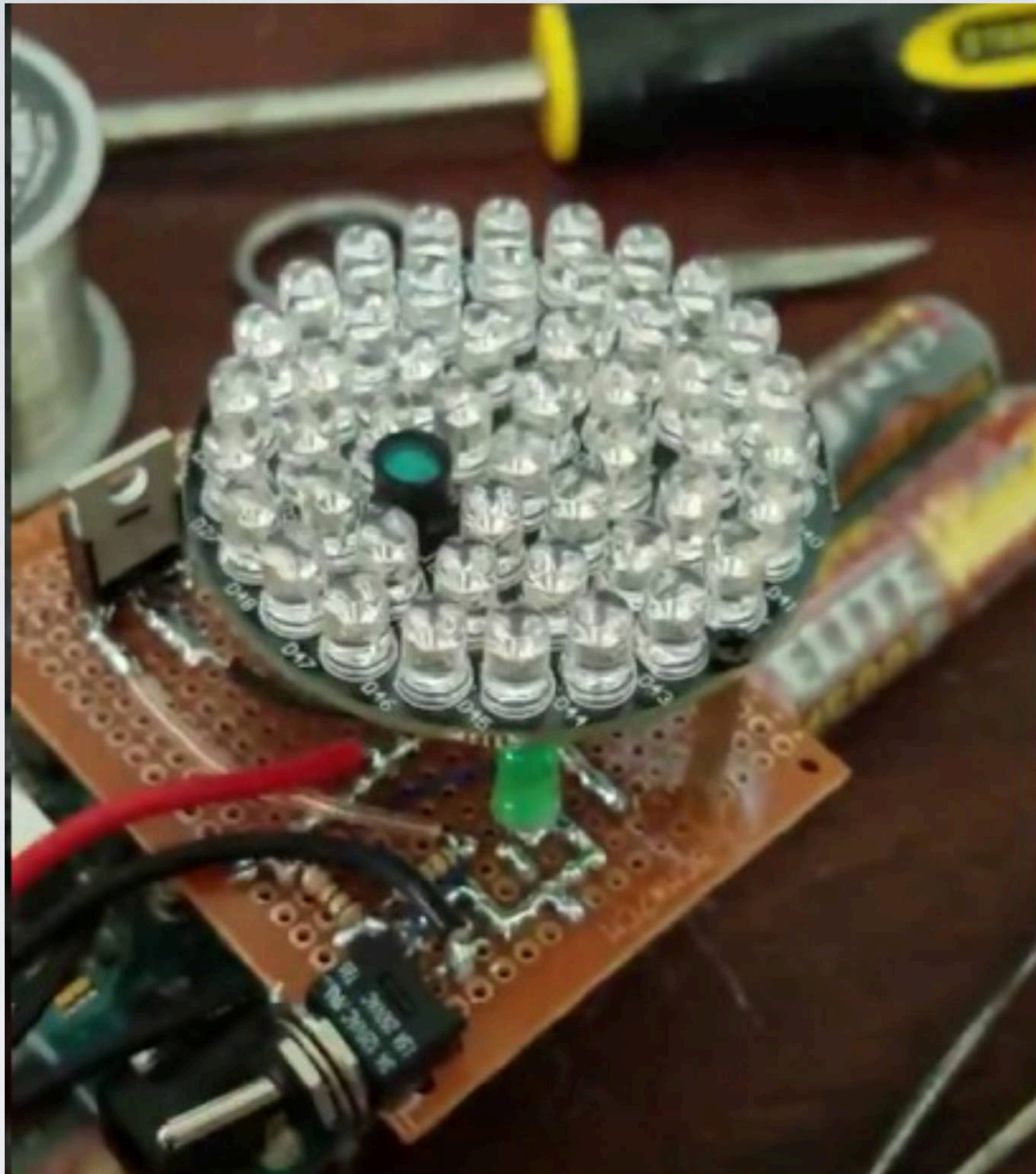
- Super Good Ideas
- Marginal Advice
- Offensive Language
- Brief Nudity

23

Slides available now at

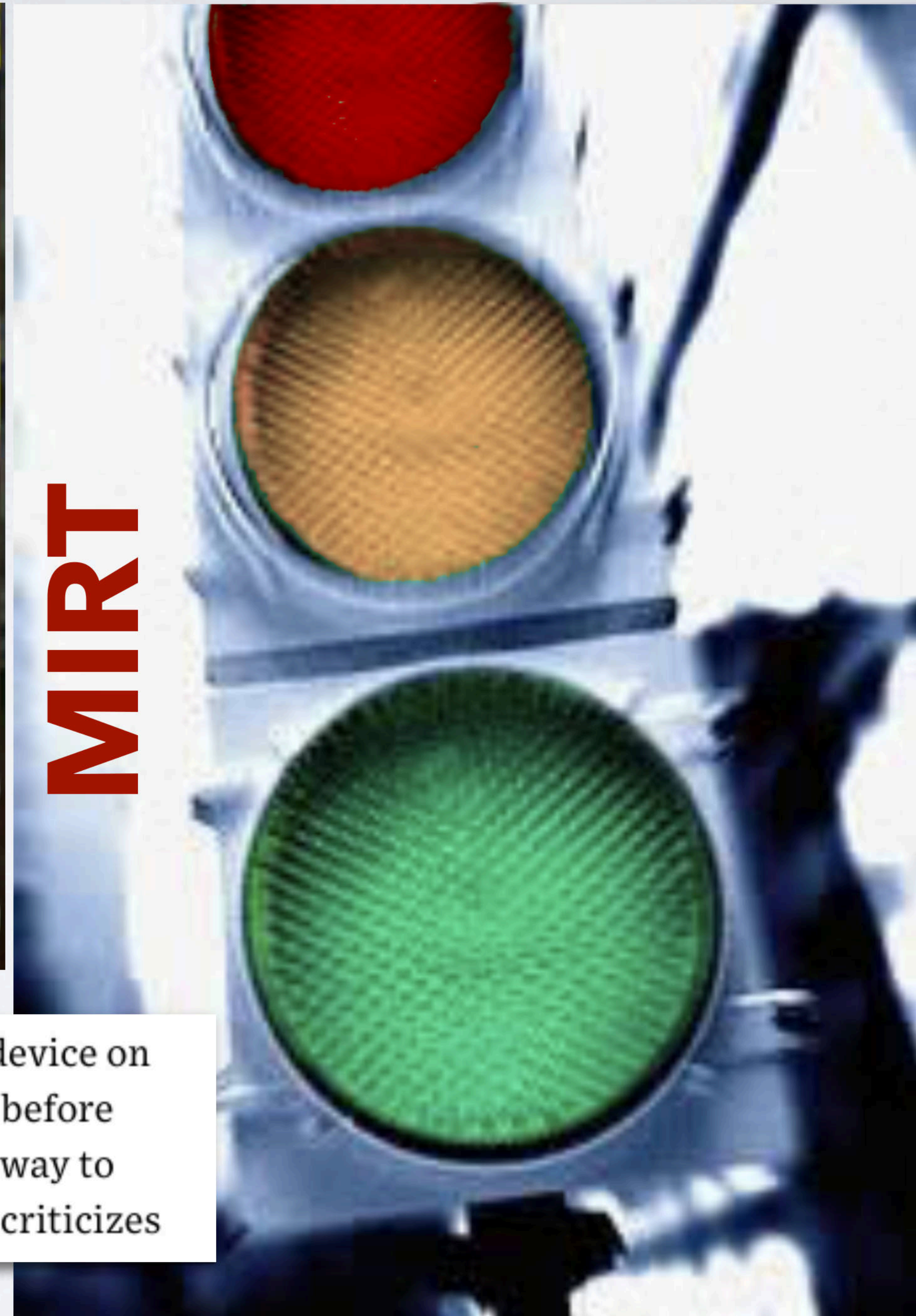


bit.ly/23police



Swearingen says he tested his proof-of-concept device on the streets of Kansas City with complete success before disassembling it to avoid trouble. "I take the freeway to work," he says. "There are no traffic signals." He criticizes

MIRT



* This is not my **MIRT**



* This is not me

HACKER

HA/KC/ER

HA/KC/ER

HA/IKER

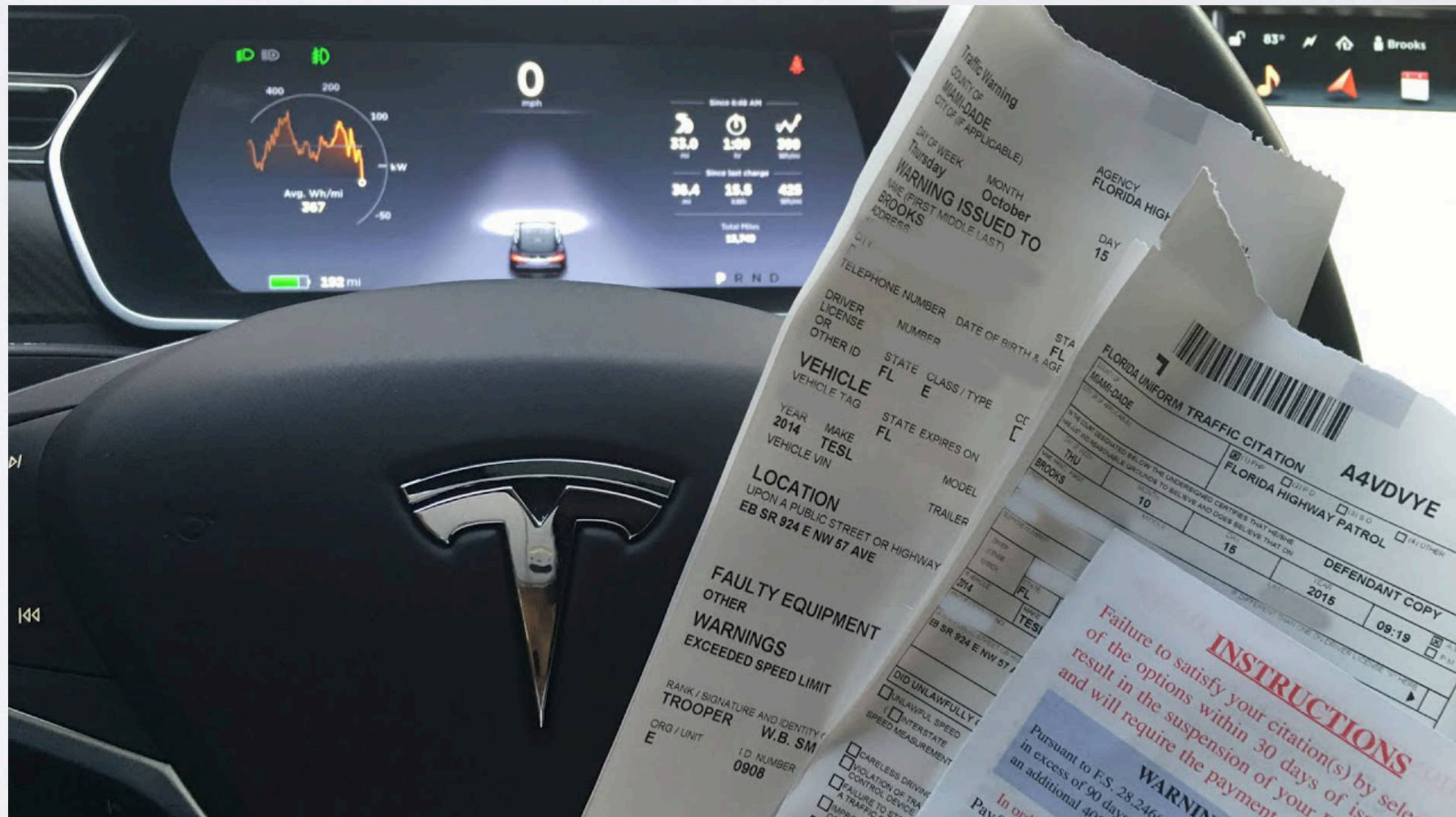
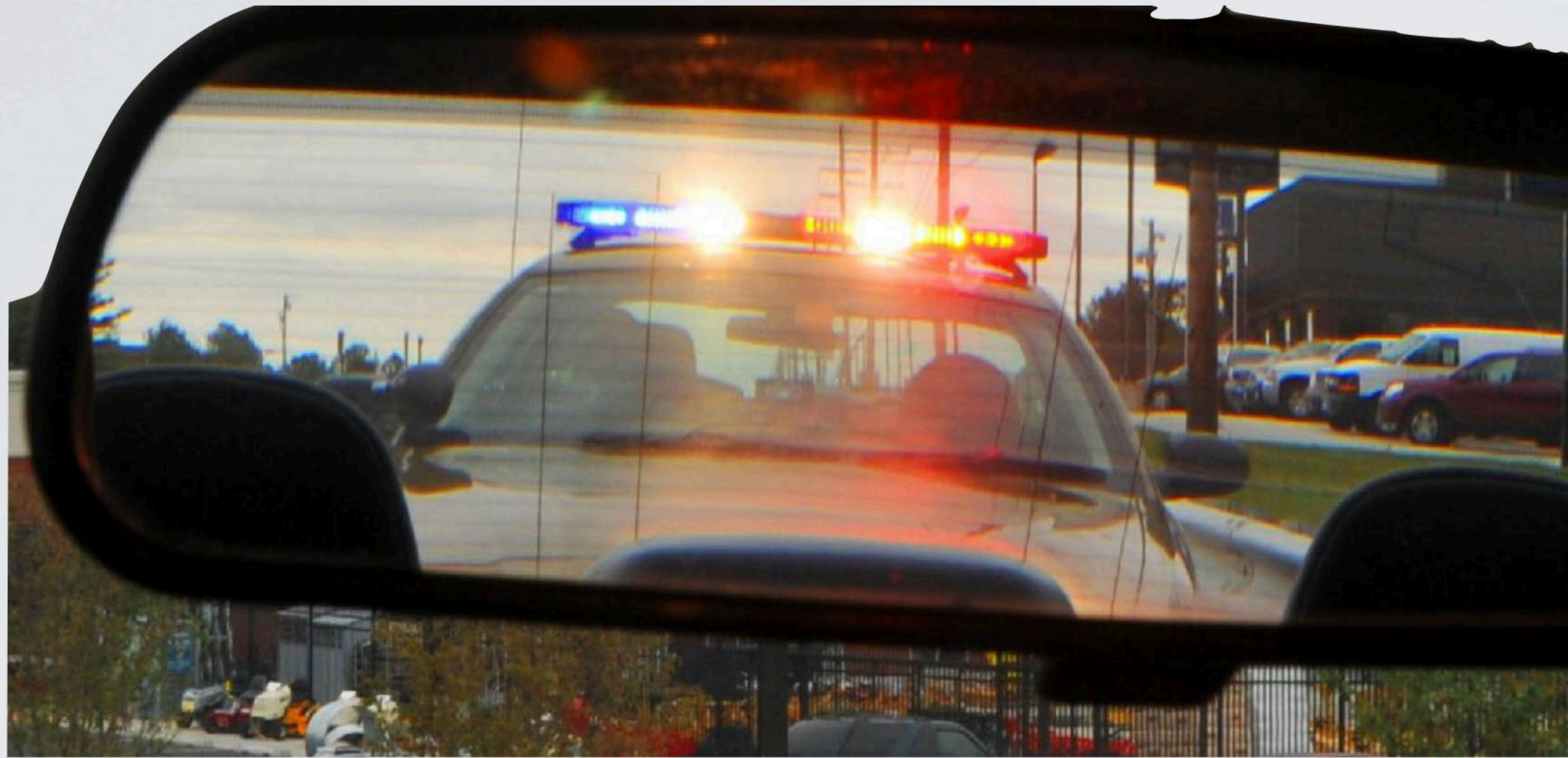


Are you a cop?





* This is not my car



* This is not my problem

WHAT IS RADAR?



IN THE 90s

MY DAD

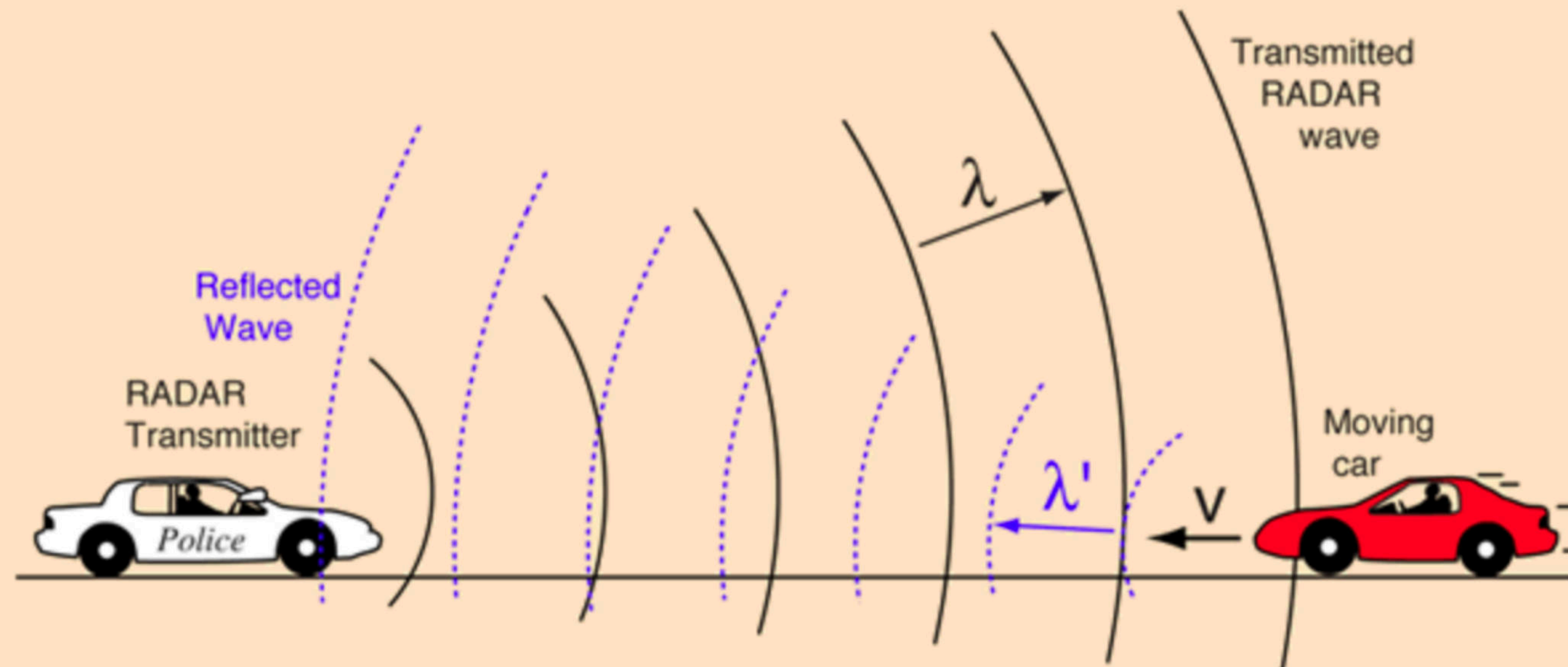
* This is not my dad

The Doppler Effect



Police RADAR

RADAR speed detectors bounce [microwave](#) radiation off of moving vehicles and detect the reflected waves. These waves are shifted in frequency by the [Doppler effect](#), and the [beat frequency](#) between the directed and reflected waves provides a measure of the vehicle speed.



[Speed detection calculation](#)

[Index](#)

[Traveling
wave
concepts](#)

[HyperPhysics](#)***** [Sound](#)

R Nave

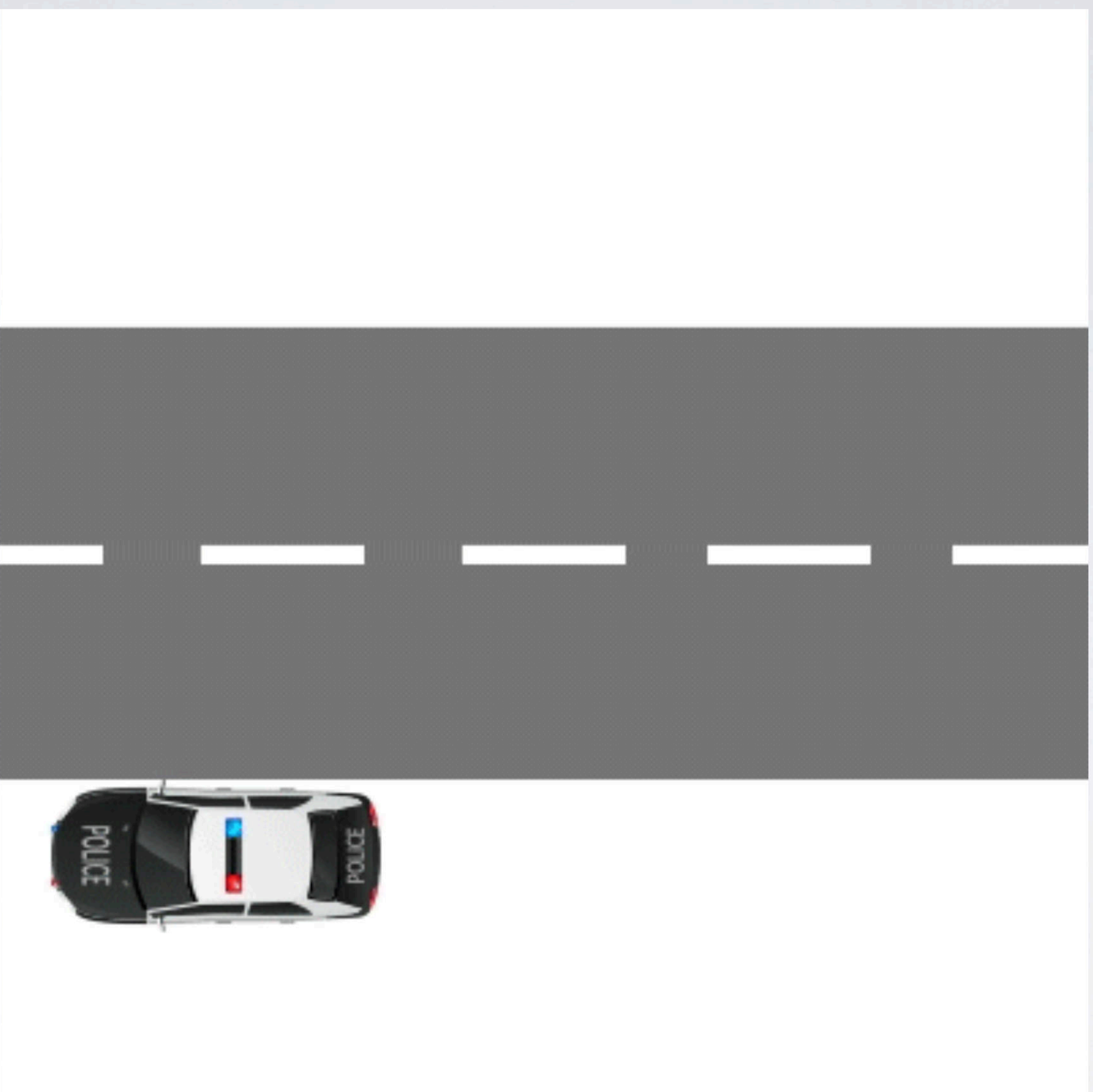
[Go Back](#)

$$\frac{\Delta f}{f} = \frac{2v_{\text{target}}}{c}$$

*Where c is the speed of light



freegifmaker.me





Police Radar Gun Frequencies

- **X Band** is in the **8.0 to 12.0 GHz** range
(most common is 10.5 in the US / 9.4-10.6 in EU)
- **K Band** is in the **18 to 26.5 GHz** range
(most common is 24.12 - 24.17 in the US / 24.12 - 25.15 in EU)
- **Ka Band** is in the **26.5 to 40 GHz** range
(most common is 33.8, 34.6, 34.7, 35.5, 35.7 GHz in the US / 34.0, 34.3, 35.5 in EU)
- **Ku Band** is in the 12 to 18 GHz range
(Ku is not used in the US, most common is 13.45 in Europe)



LIVE



DEMO



Police Radar Gun Frequencies

- **X Band** is in the **8.0 to 12.0 GHz** range
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(Ku is not used in the US, most common is 13.45 in Europe)

From: <https://www.vortexradar.com>



$$\frac{\Delta f}{f} = \frac{2v_{\text{target}}}{c}$$

Defeat X band radar at

10.50000203125Ghz

Defeat K band radar at

24.12000466607143Ghz

Confuse? Ka band radar at

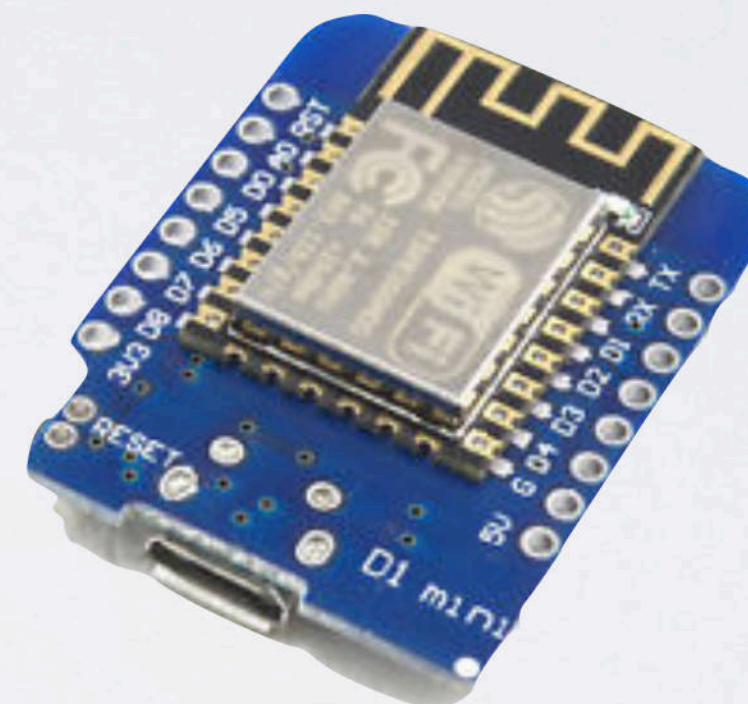
33.80000653869047Ghz

Microwave Oven 2.4Ghz = -97,420,118 mph

<http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/radar.html>



+



**WE HAVE
A PROBLEM**

good idea meter: RED

WE HAVE A PROBLEM



good idea meter: RED



Google Maps Platform

Web Services > Roads API



Google Maps Platform

Web Services > Roads API

**The Roads API
returns the
posted speed
limit for a given
road segment.**

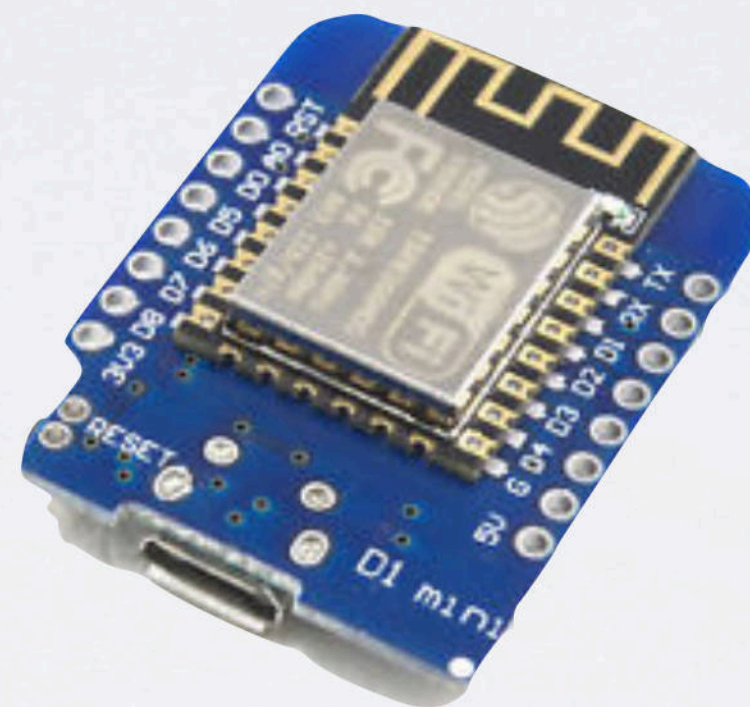
Response

```
{
  speedLimits:
  [
    {
      placeId: "ChIJX12duJAwGQ0Ra0d40i4j0GE",
      speedLimit: 105,
      units: "KPH"
    },
    {
      placeId: "ChIJLQcticc0GQ0RoiNZJVa5GxU",
      speedLimit: 70,
      units: "KPH"
    },
    {
      placeId: "ChIJJ4vQRudkJA0RpednU70A-5M",
      speedLimit: 55,
      units: "KPH"
    }
  ],
}
```






+



+



good idea meter: RED



PUBLIC NOTICE

FEDERAL COMMUNICATIONS COMMISSION
1919 M STREET, N.W.
WASHINGTON, D.C. 20554

DA 96-2040

News media information 202/418-0500 Fax-On-Demand 202/418-2830 Internet: <http://www.fcc.gov> <ftp.fcc.gov>

Released: December 9, 1996

FCC REGULATES RADAR TRANSMITTERS, BUT NOT RADAR DETECTORS

**Using a radar jammer is a
federal offense that can
result in a \$50,000 fine
and/or five years in prison.**



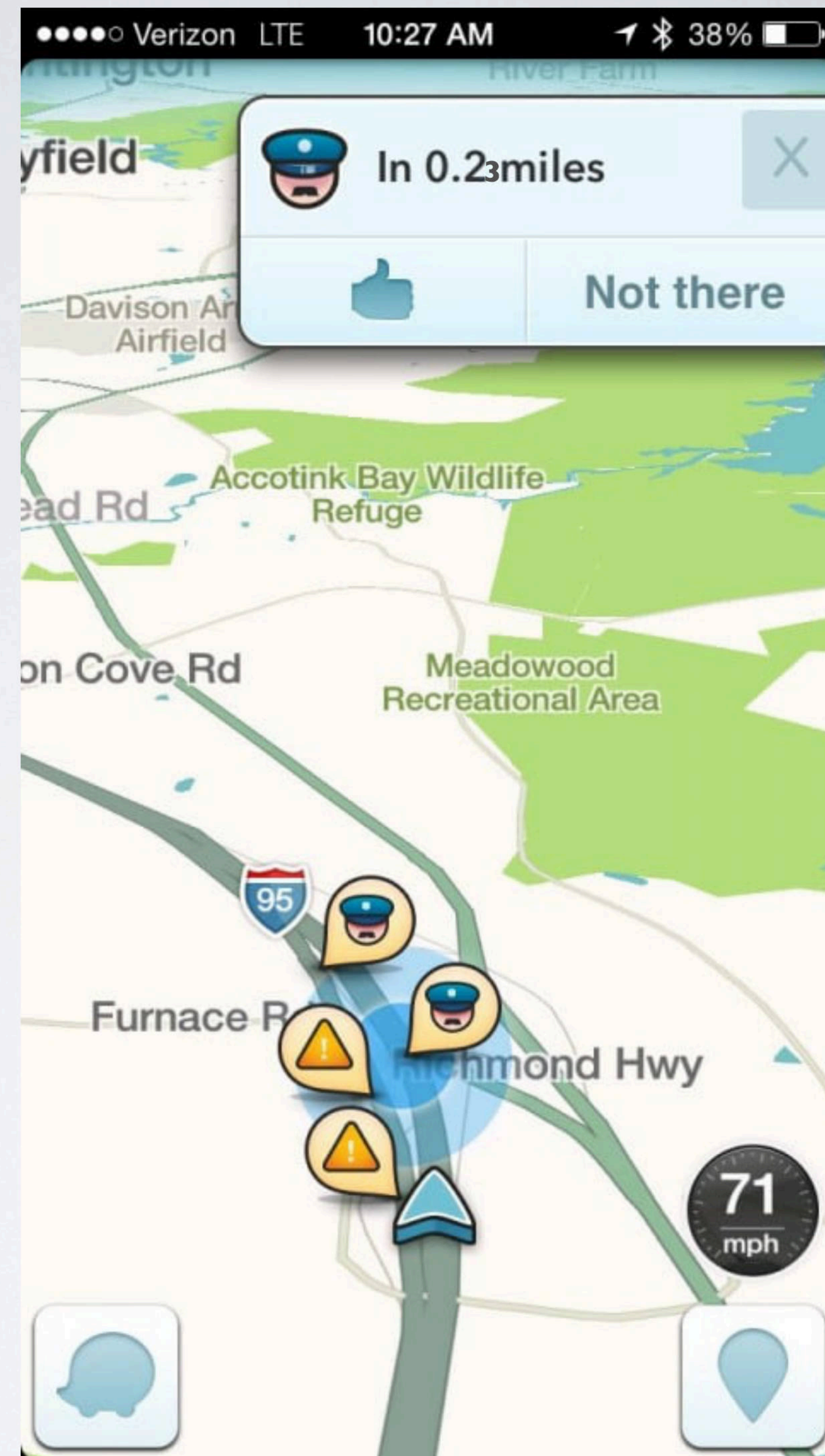
* This is still not me

(LEGAL) EFFECTIVE COUNTERMEASURES

EFFECTIVE COUNTERMEASURES



- Uniden R3 or R7
- Escort Max360
- Radenso Pro M
- Valentine One w/BT





LASER

**"The FCC doesn't
regulate the light
spectrum. That's done
by the Food and Drug
Administration"**

—autoblog.com, 2011

LOFT



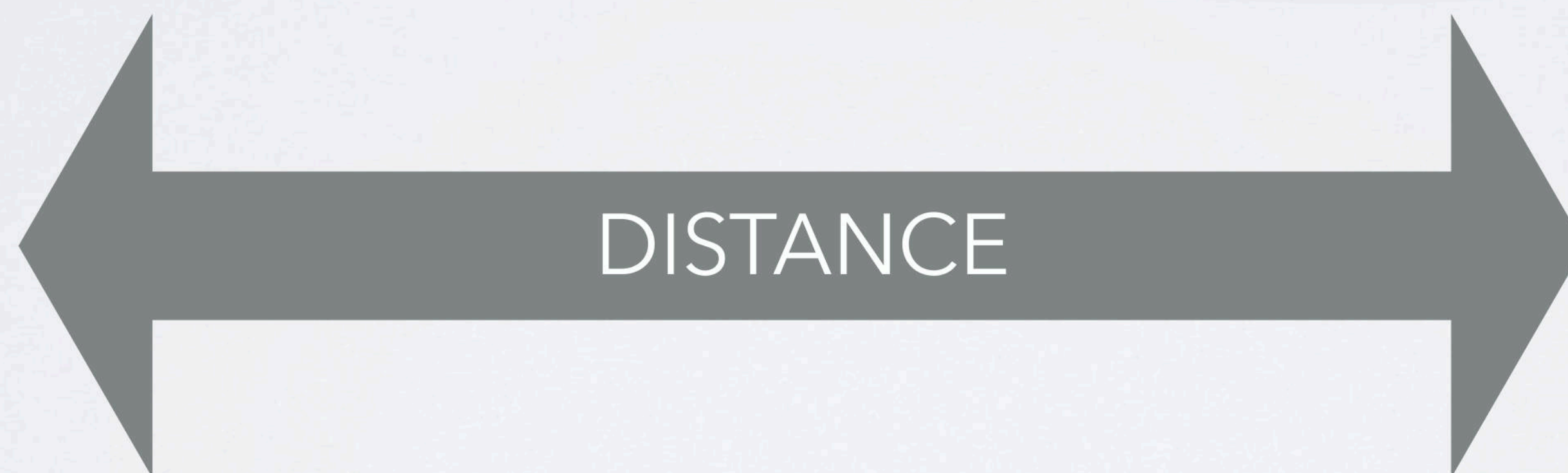


*** This is not my car either**

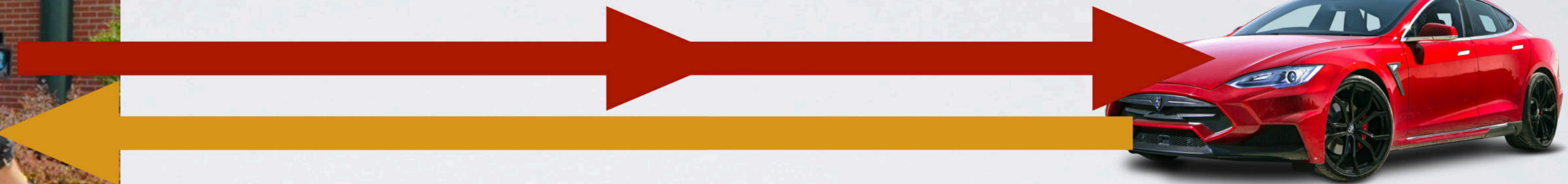




CLASS 1 LASER
904nm (Invisible IR)



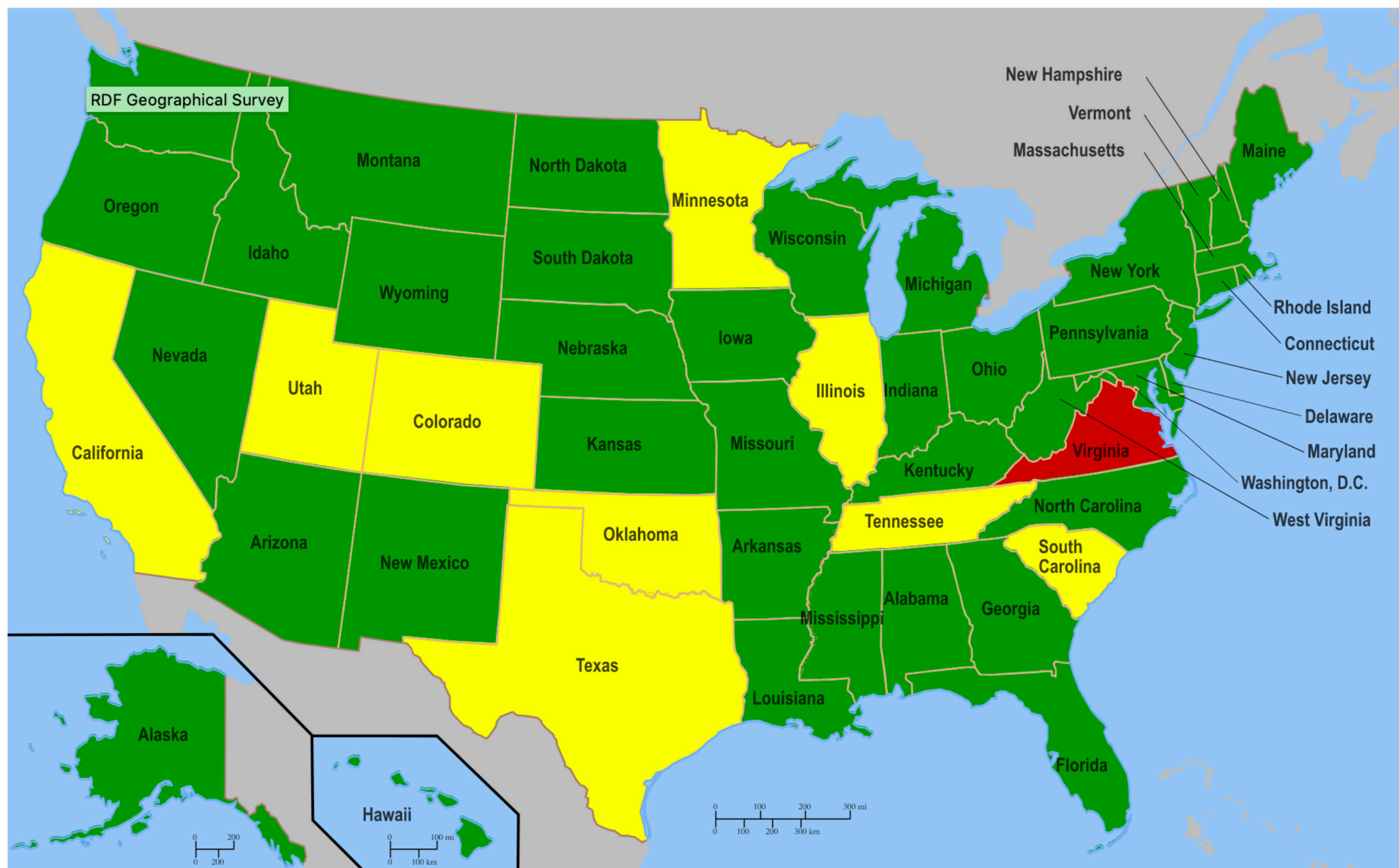
$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$



$$\text{Distance} = \frac{C * T}{2}$$

(MOSTLY-LEGAL) EFFECTIVE COUNTERMEASURES

good idea meter: **IDK I AM A METER, NOT A COP**



Legend:

RED: ALL COUNTERMEASURES ILLEGAL

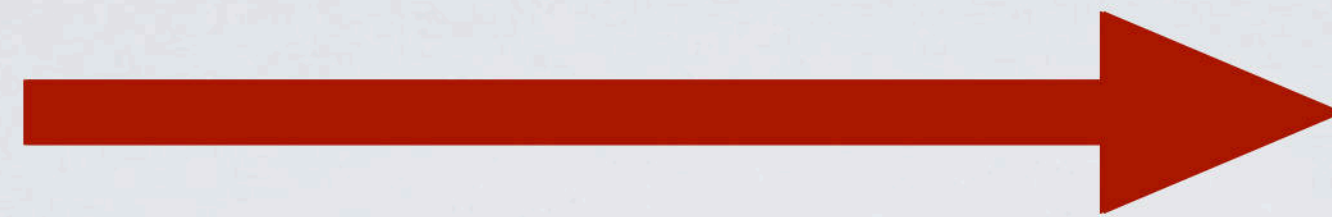
YELLOW: LASER JAMMING ILLEGAL

GREEN: ROCK & ROLL

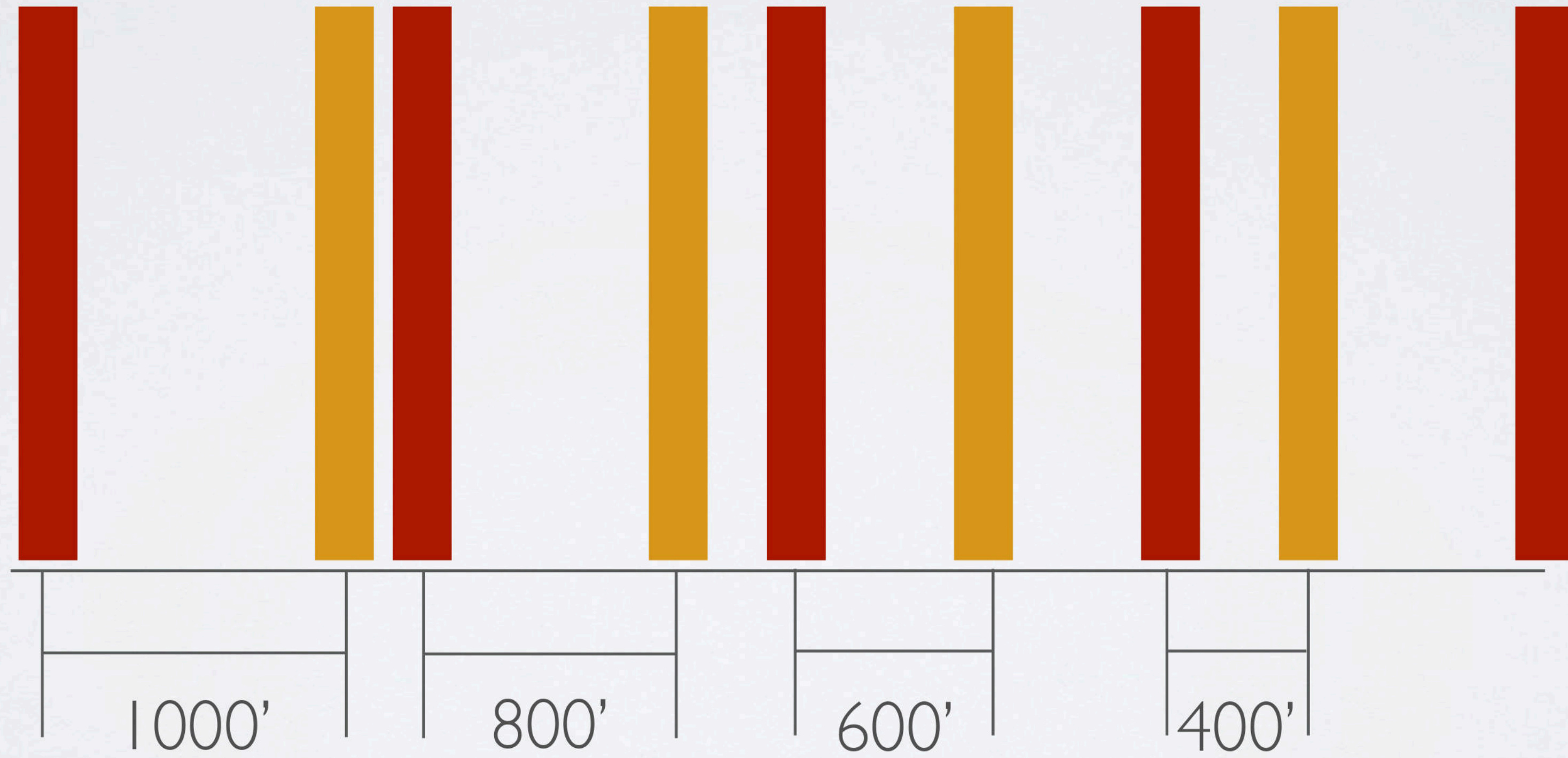
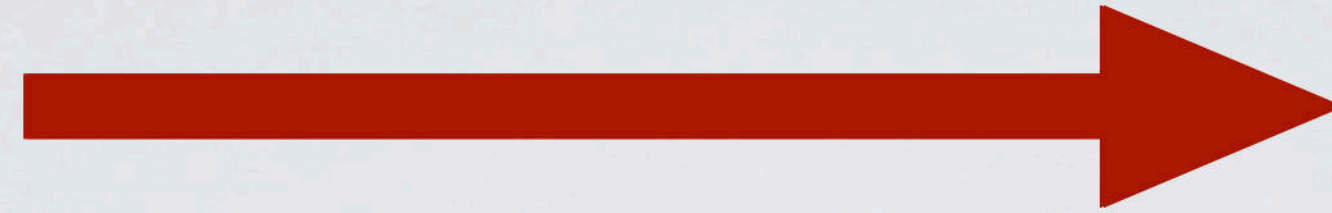
*Luckily nobody is here from the DC Area

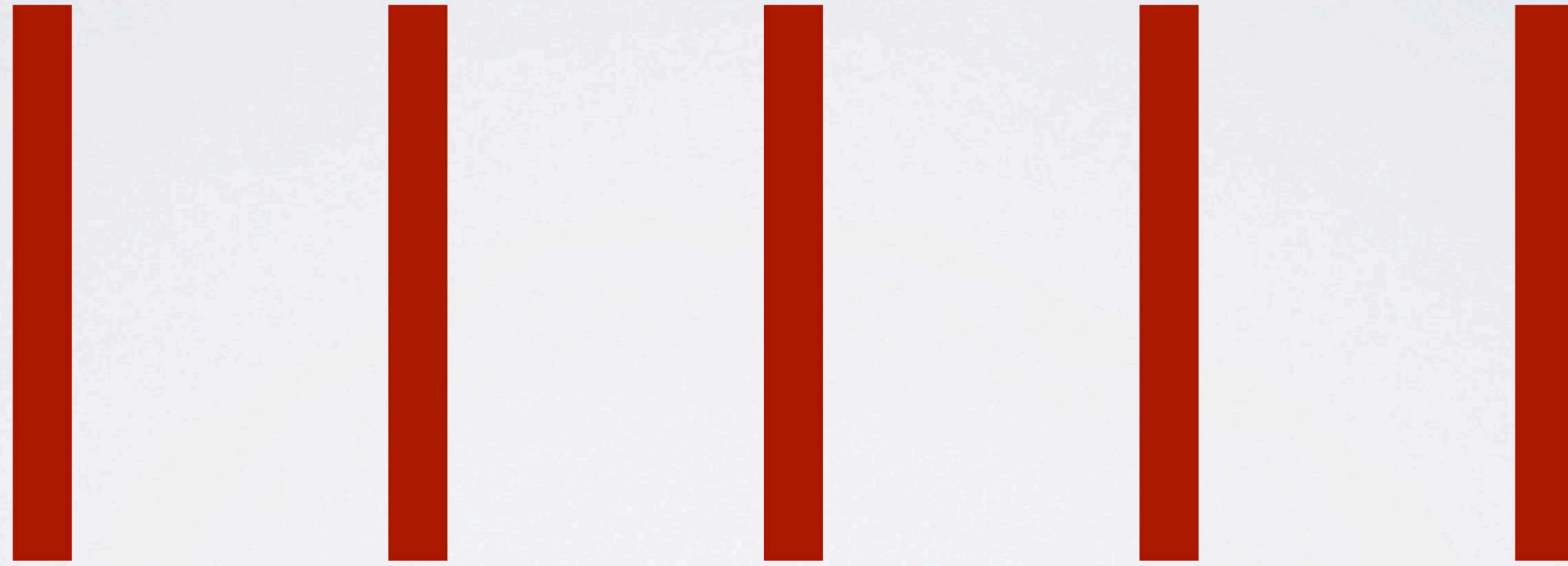
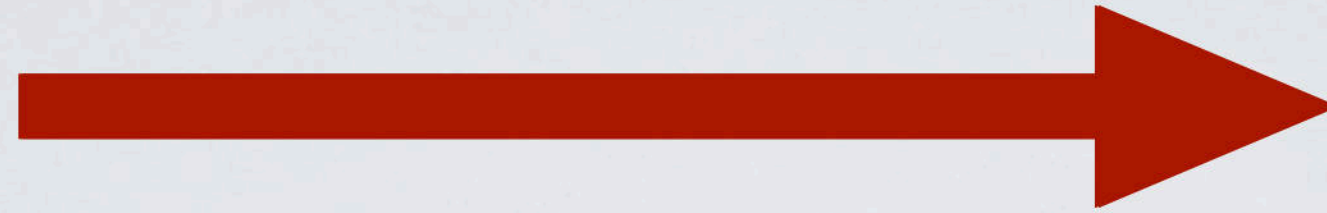
OPTION 1



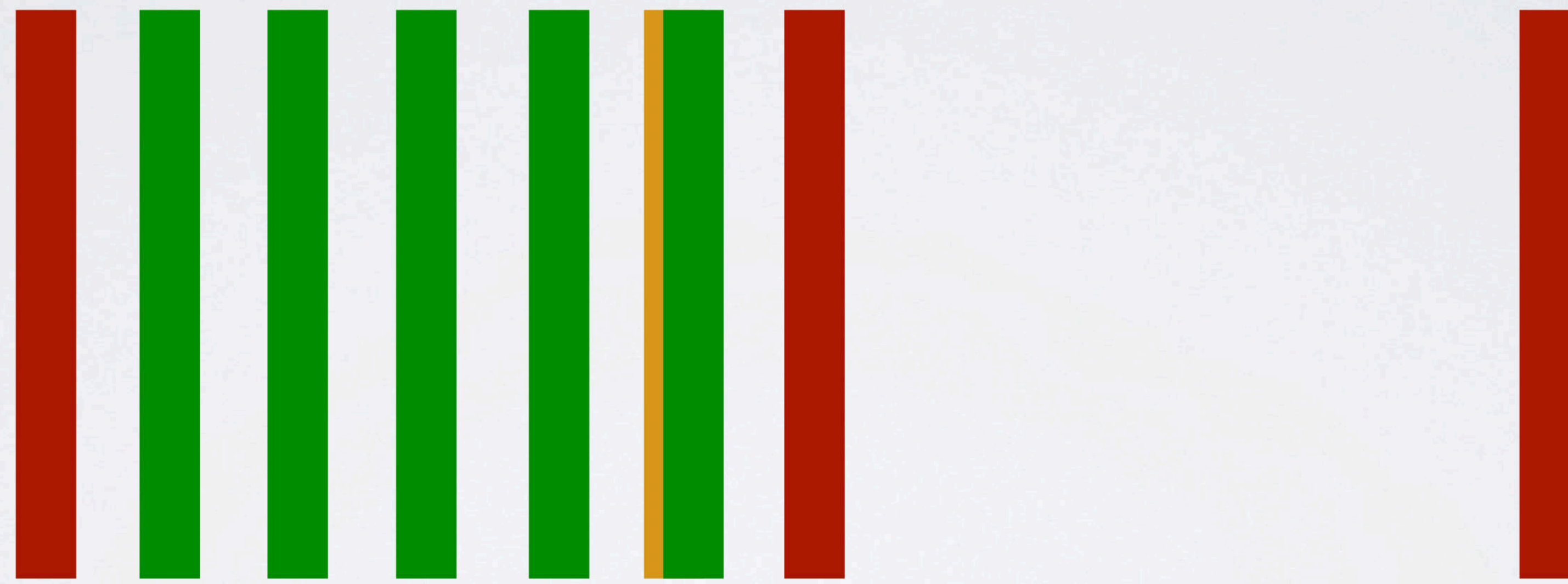


5ms

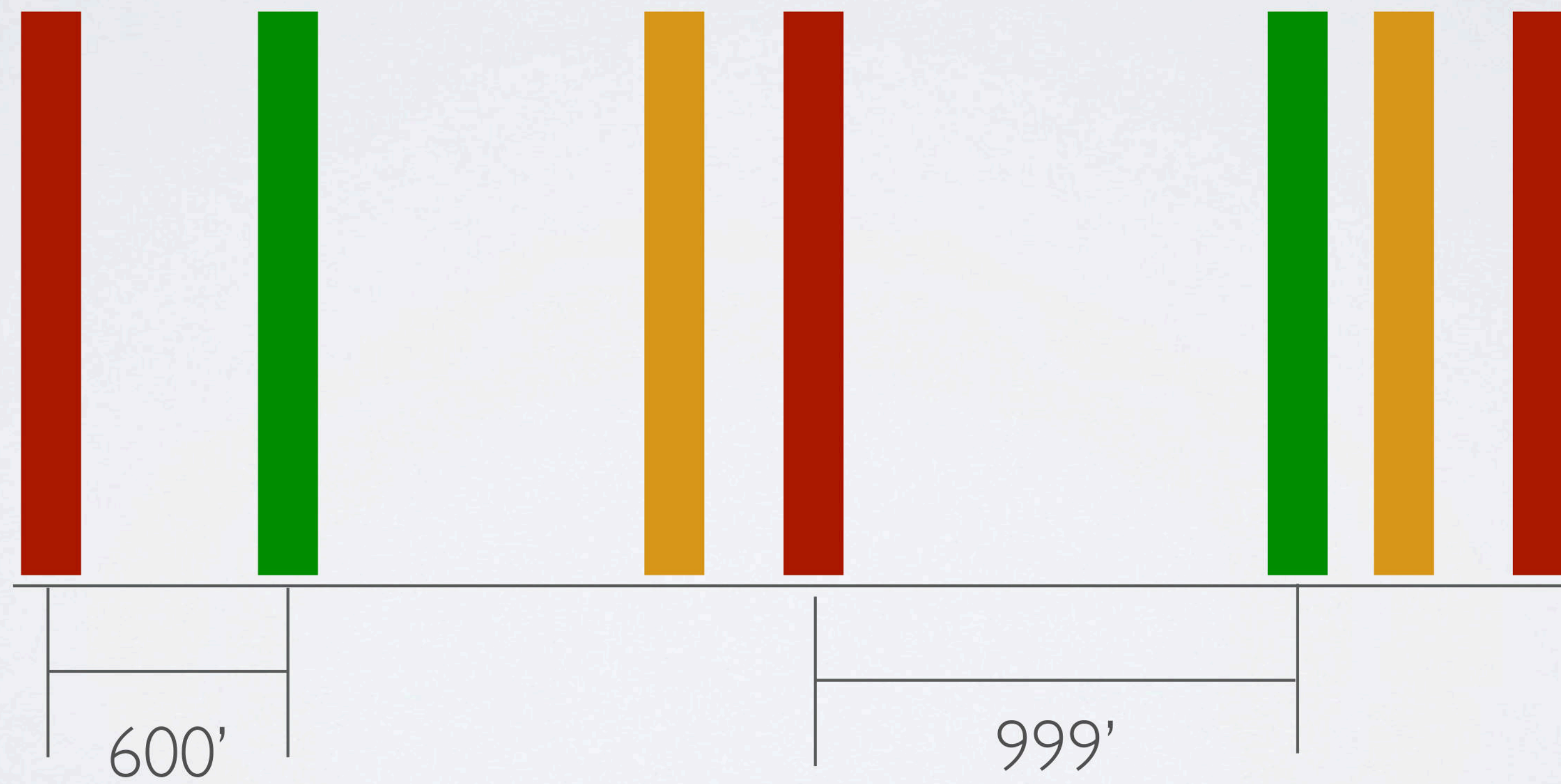


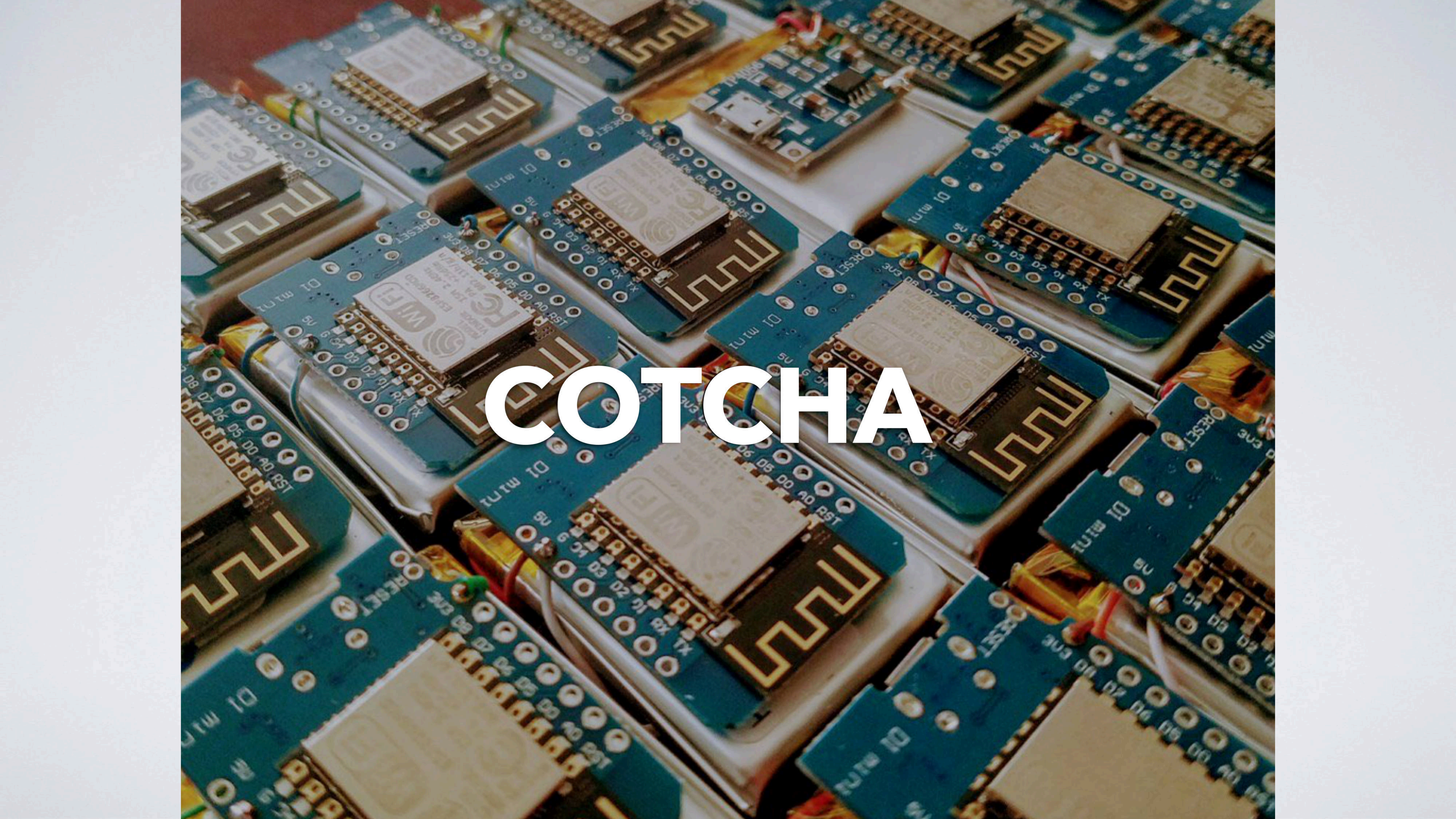


5ms



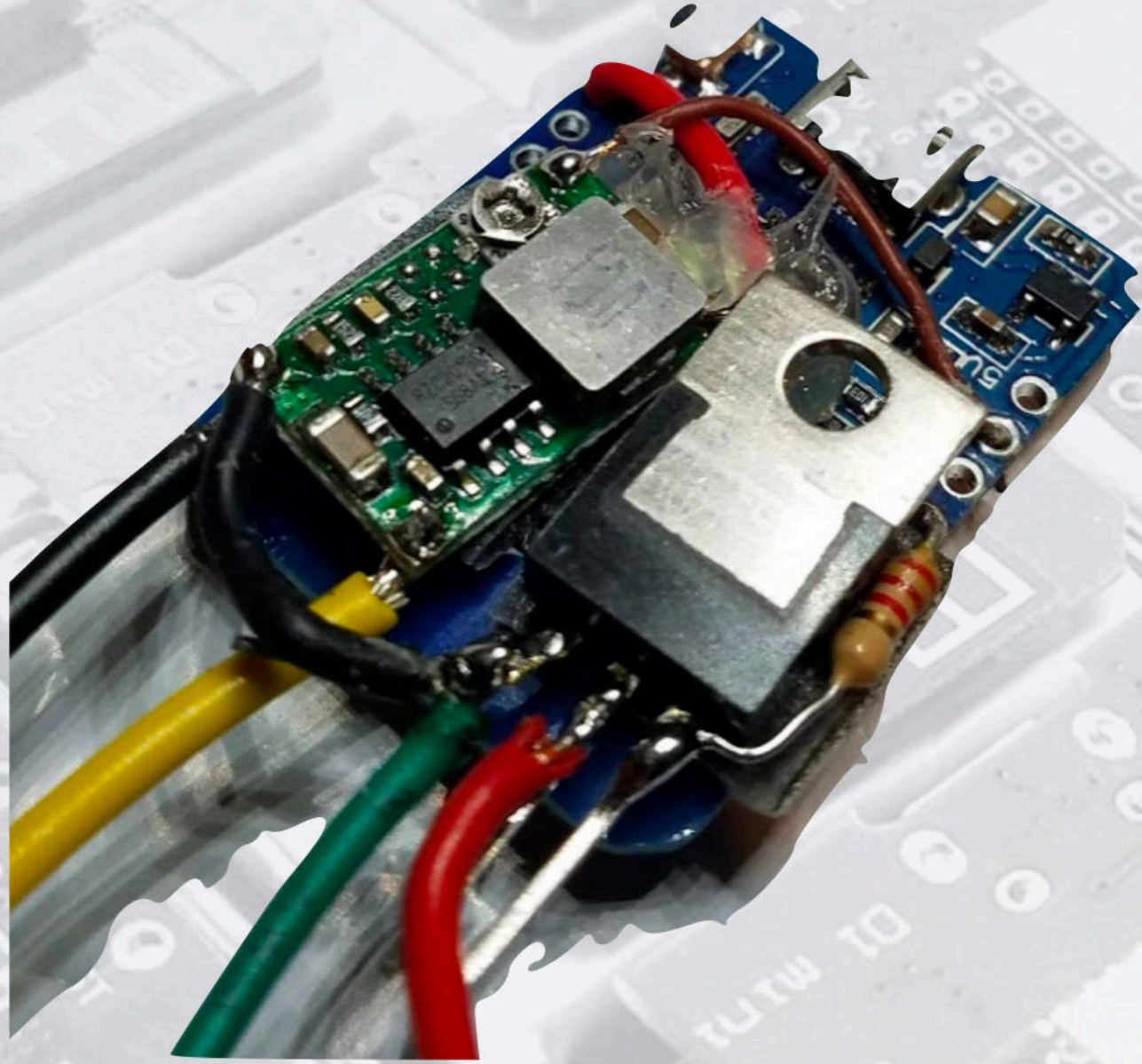
5ms





СОТЧНА

NOTCHACOTCHA



- ESP8266 Based Laser Jammer
- 12V Circuitry for Automobile Install
- 940nm Brute Force Mode
- Wireless Connectivity & Android App
- **Legal in SOME STATES!**
- Laser Gun Emulation Mode
- ~~MIRT Mode~~ (good idea meter: **SUPER RED**)

NOTCHACOTCHA

IS

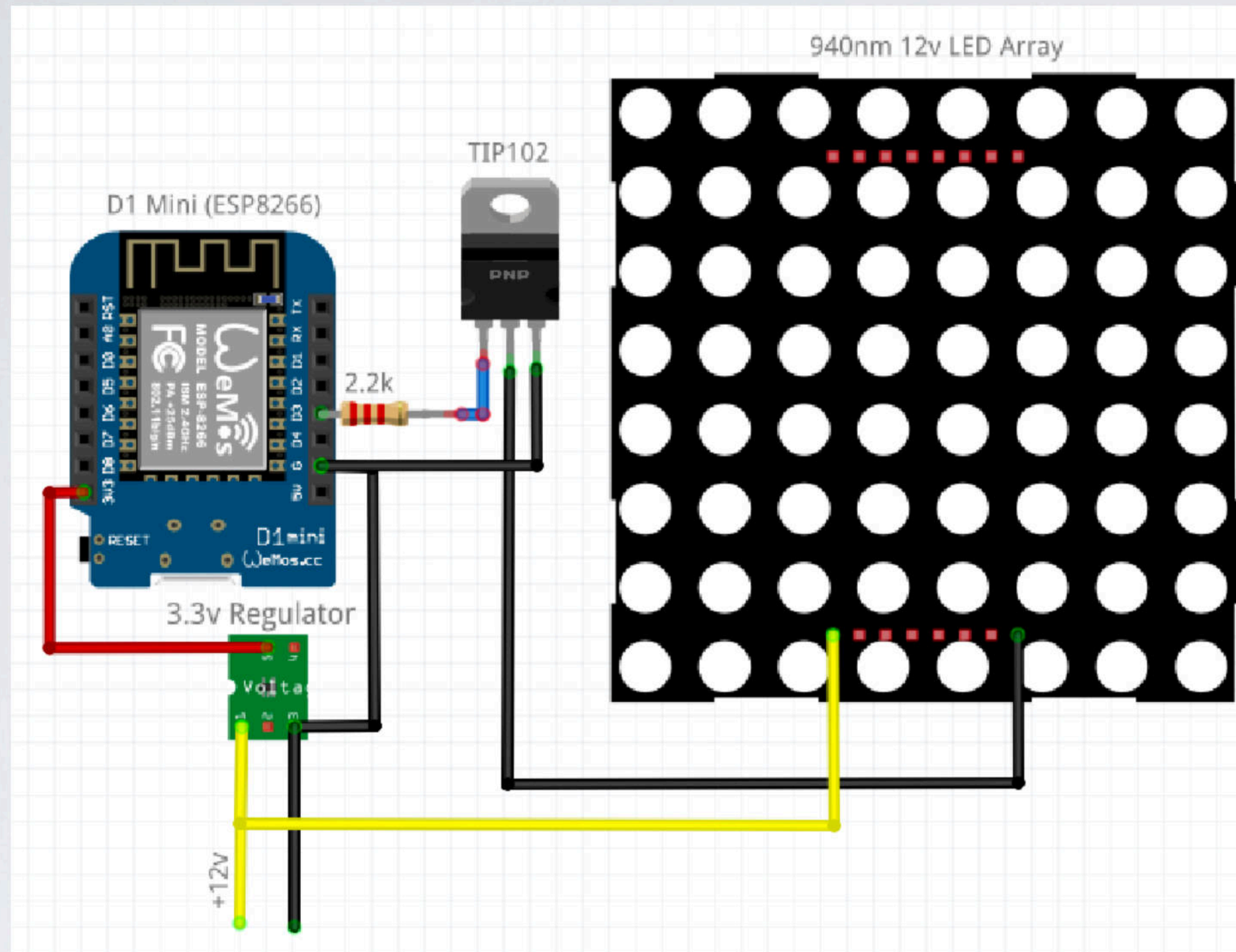
FREEDOM

AMERICA!



*EXCEPT VIRGINIA

NOTCHACOTCHA



- D1 Mini (esp8266) : \$1.50
- 2.2k Resistor : \$0.03
- 3.3v Voltage Regulator : \$0.54
- TIP102 Transistor : \$0.08
- 940nm LED Array : \$6.00

\$8.15

**bill of materials, code, and several other
bad ideas at github.com/hevnsnt/NOTCHACOTCHA**

This is only round one

Bill Swearingen / &bill / @hevnsnt

Slides available now

bit.ly/23police

Super duper thanks to:

@VortexRadar

thejunkyardmessiah

www.rdforum.org

[kickass](#)

@sec_kc / @networkgeek / @seckcory / @surbo

~#RexKC sucks and #23isReal~